

Three-Player Bilevel Generalized Nash Benchmark

Problem Statement

This document writes the three-player bilevel benchmark used in the test suite. There are three players, indexed by $i \in \{1, 2, 3\}$. Player i 's decision vector is

$$x_i = (x_{i,1}, x_{i,2}, x_{i,3}, x_{i,4}, x_{i,5}) \in \mathbb{R}^5,$$

where $x_{i,j}$ denotes the j -th decision variable of player i . In the code this is 'x[Block(i)][j]'.
The bilevel preference order is:

upper level controls coordinates $\{4, 5\}$, lower level controls coordinates $\{1, 2, 3\}$.

Raw Objective Targets

The raw objective target is the point each player's objective would choose if there were no constraints. The constraints then force the final equilibrium away from this raw target.

The known equilibrium is

$$\begin{aligned} x_1^* &= (0.40, 0.90, 0.55, 1.20, 0.80), \\ x_2^* &= (0.45, 1.00, 0.60, 1.30, 0.90), \\ x_3^* &= (0.50, 1.10, 0.65, 1.40, 1.00). \end{aligned}$$

The raw target for player i is

$$t_i = x_i^* + (0.6, 0, -0.4, 0, 0).$$

Thus

$$\begin{aligned} t_1 &= (1.00, 0.90, 0.15, 1.20, 0.80), \\ t_2 &= (1.05, 1.00, 0.20, 1.30, 0.90), \\ t_3 &= (1.10, 1.10, 0.25, 1.40, 1.00). \end{aligned}$$

Quadratic Bilevel Problem

For each player i , define the upper-level and lower-level objectives

$$J_i^{\text{up}}(x_i) = \sum_{j \in \{4,5\}} (x_{i,j} - t_{i,j})^2,$$

and

$$J_i^{\text{low}}(y_i) = \sum_{j \in \{1,2,3\}} (y_{i,j} - t_{i,j})^2.$$

The bilevel problem for player i is

$$\begin{aligned} \min_{x_i \in \mathbb{R}^5} \quad & J_i^{\text{up}}(x_i) \\ \text{subject to} \quad & x_i \in \arg \min_{y_i \in \mathbb{R}^5} \left\{ J_i^{\text{low}}(y_i) : g_i(y_i, x_{-i}) \geq 0 \right\}, \end{aligned}$$

where x_{-i} denotes the decision vectors of the other two players.

Coupled Generalized Nash Constraints

Let

$$\rho = 0.25$$

and define the coupled resource usage

$$c_{i,j}(x) = x_{i,j} + \rho \sum_{k \neq i} x_{k,j}.$$

Here $c_{i,j}(x)$ is a function of the current players' decisions. The uppercase constants below are fixed capacities or floors.

Each player has four coupled inequalities:

$$g_i(x) = \begin{bmatrix} C_{i,1} - c_{i,1}(x) \\ C_{i,2} - c_{i,2}(x) \\ c_{i,3}(x) - F_{i,3} \\ c_{i,4}(x) - F_{i,4} \end{bmatrix} \geq 0.$$

The constants are chosen from the known solution x^* :

$$\begin{aligned} C_{i,1} &= c_{i,1}(x^*), \\ C_{i,2} &= c_{i,2}(x^*) + 1, \\ F_{i,3} &= c_{i,3}(x^*), \\ F_{i,4} &= c_{i,4}(x^*) - 1. \end{aligned}$$

Numerically,

i	$C_{i,1}$	$C_{i,2}$	$F_{i,3}$	$F_{i,4}$
1	0.6375	2.4250	0.8625	0.8750
2	0.6750	2.5000	0.9000	0.9500
3	0.7125	2.5750	0.9375	1.0250

Expected Solution and Active Set

At the expected equilibrium x^* ,

$$C_{i,1} - c_{i,1}(x^*) = 0, \quad c_{i,3}(x^*) - F_{i,3} = 0.$$

Thus the active constraints for each player are:

$$C_{i,1} - c_{i,1}(x) \geq 0, \quad c_{i,3}(x) - F_{i,3} \geq 0.$$

The inactive constraints have slack equal to 1:

$$C_{i,2} - c_{i,2}(x^*) = 1, \quad c_{i,4}(x^*) - F_{i,4} = 1.$$

The constraints are genuinely coupled because every $c_{i,j}(x)$ depends on all three players' decision variables. The active coordinate-1 and coordinate-3 resource equations are player-specific, so they pin the distribution across players rather than only aggregate totals.

Nonlinear Variant Used in Tests

The nonlinear test case keeps the same known solution and active set. It replaces the squared-distance objective by

$$\sum_{j \in S} (\cosh(x_{i,j} - t_{i,j}) - 1),$$

and replaces each linear inequality residual $r(x) \geq 0$ by

$$\exp(r(x)) - 1 \geq 0.$$

This preserves the feasible set and active set because

$$\exp(r) - 1 = 0 \iff r = 0, \quad \exp(r) - 1 > 0 \iff r > 0.$$